



Physics with Year Abroad, Literature Review: Particle Identification in Modern Cherenkov Imaging Detectors

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With major detector assemblies lining up for construction at high-energy facilities such as EIC and FAIR, technologies for particle identification are becoming ever more relevant. This review motivates the theory of the field, introduces the technologies behind modern Cherenkov imaging detectors, and discusses the implementation of these systems into upcoming experiments. Emphasis is placed on event reconstruction techniques and the challenges presented by cutting-edge physics programmes. Persistent challenges include the continued development of photosensitive devices and deploying modern software and electronics to detector systems. We conclude by outlining promising future developments in the field.

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21 I. INTRODUCTION

22 Modern high energy physics experiments rely on the
23 performance of the detector systems to accurately study
24 collision events. Cherenkov imaging detectors will play
25 a central role in the upcoming generation of experiments
26 in quantum chromodynamics, gluon research and high-
27 energy neutrinos. Improving the capabilities of these

28 imaging detectors is the work of thousands of physicists
29 and engineers around the world.

1 30 This review aims to provide an introduction to the
31 field of Cherenkov specific systems for particle identi-
1 32 fication (PID). We will cover the basic principles of
1 33 Cherenkov radiation, some design choices of Cherenkov
2 34 imaging detectors, and briefly over the latest advance-
35 ments in technology. Cherenkov imaging detectors can
2 36 be broadly divided into angle-imaging systems, such as
37 RICH and DIRC detectors used for PID at accelerators,
3 38 and topology-imaging systems, such as large-volume wa-
3 39 ter or ice Cherenkov detectors employed in neutrino and
4 40 astroparticle physics.

4 41 We will begin with an overview of the theoretical back-
4 42 ground in Sect II, and then introduce photomultipliers in
5 43 Sect III. Following this, we will delve into the main sub-
44 classes of Cherenkov imaging detectors in Sect IV, where
6 45 over the chapters we will discuss their designs and dif-
46 ferences. Finally, we will conclude with a discussion of
7 47 some open research in Sect V.

8 II. THEORETICAL FOUNDATIONS

8 II.A. Cherenkov Radiation

50 Cherenkov radiation is a phenomenon that occurs
when a charged particle travels through a dielectric
medium with refractive index $n(\lambda)$ at a speed greater
than the phase velocity in that medium. This effect was
first observed by Pavel Cherenkov in 1934¹ and later ex-
plained theoretically by Ilya Frank and Igor Tamm².

The angle of Cherenkov radiation θ_c relative to the

particle track, for a particle of speed βc in a medium with refractive index n is

$$\cos \theta_c = \frac{1}{n\beta}. \quad (1)$$

From Maxwell's equations arises that electromagnetic propagation with wavelength λ will have its phase velocity modified by the medium it is travelling through. This leads to the idea of refractive index $n(\lambda) = c/v_p$, defined as the ratio between the speed of light in vacuum c and the phase velocity v_p in the medium. The condition for Cherenkov radiation to occur is then $v > c/n$.

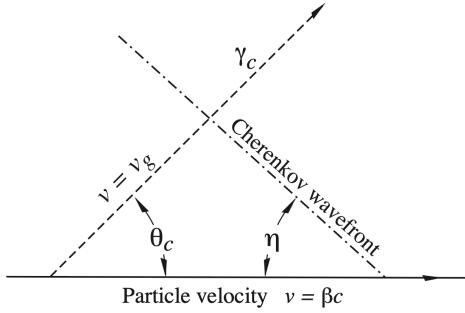


FIG. 1 **Cherenkov emission of photons and wavefront.** A particle travels through a medium with velocity v . Photons are emitted at the group velocity v_g and angle θ_c to the path of travel of the particle, giving the characteristic blue shock cone. In a dispersive medium, $\theta_c + \eta \neq 90^\circ$ generally³.

Let the photons have frequency ω and wavenumber $k = 2\pi/\lambda$. They propagate at the group velocity, $v_g = d\omega/dk$. The photons are concentrated in a thin conical shell as shown in Fig 1, with an opening half-angle η as

$$\cot \eta = \left[\tan \theta_c + \beta^2 \omega n(\omega) \frac{dn}{d\omega} \cot \theta_c \right] d\omega \quad (2)$$

where we are considering a small frequency range. Unless the medium is non-dispersive, where $dn/d\omega = 0$, generally $\theta_c + \eta \neq 90^\circ$, and the Cherenkov wavefront 'sideslips' along with the particle³.

The number of photons emitted per unit length is given by the Frank-Tamm formula

$$\frac{d^2 N}{dx d\lambda} = \frac{2\pi\alpha z^2}{\lambda^2} \left(1 - \frac{1}{\beta^2 n^2(\lambda)} \right) \quad (3)$$

for a particle with charge ze and wavelength λ , moving through a medium with refractive index $n(\lambda)$, and where α is the fine structure constant². For most uses, Equ 3 should be integrated over the region $\beta n(\lambda) > 1$. The number of photons N emitted per unit length is inversely proportional to the square of the wavelength, explaining the characteristic blue glow of Cherenkov radiation.

II.B. Mass Spectrometry

Mass spectrometry refers to the mass measurement of a particle. There are many different methods, such as time-of-flight (TOF), gas chromatography, quadrupole

ion traps, etc. Mass spectrometry in Cherenkov imaging detectors principally requires the measurement of the Cherenkov angle of a particle at a known momentum. Using the relativistic transformations for momentum and energy

$$p = \gamma m_0 v \quad (4)$$

$$E = \gamma m_0 c^2 \quad (5)$$

we can rewrite the Cherenkov angle from Equ 1 as

$$\cos \theta_c = \frac{\sqrt{p^2 + m^2 c^2}}{np}. \quad (6)$$

The Cherenkov angle and the momentum can be used to infer the mass of the particle in a known medium. This equation can be used to plot the analytical curves for the experimental data in Fig 2.

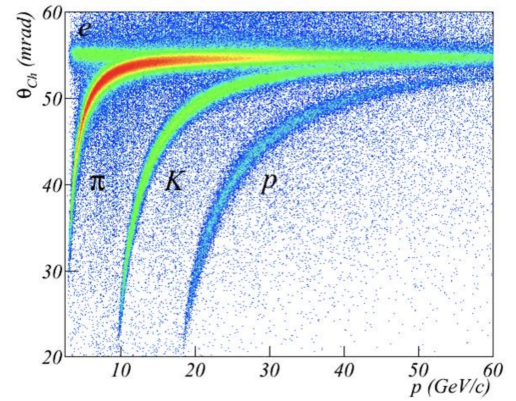


FIG. 2 **Cherenkov spectrometry of the COMPASS RICH-1 detector at CERN⁴.** There are clear curves for different particle species. The analytical form of these curves can be determined using Equ 6. Measurements of points on this curve can provide unique PID.

In modern detector arrays at collider setups, forward detectors provide momentum measurements using TOF systems. Then, Cherenkov angle-imaging detectors (such as RICHs in Sect IV.A) provide measurement of the Cherenkov angle by reconstructing the cone through analysis of the Cherenkov photon hits incident on a photodetector array. Topology-imaging detectors (such as Super-K in Sect IV.C) map the entire Cherenkov cone of photons using photodetectors over a huge area, and can obtain momentum estimates from the light intensity.

II.C. Statistical Foundations of Event Reconstruction

The problem of event reconstruction in modern Cherenkov detectors is to determine what happened inside the detector based on the data obtained from the photon sensors. The most common current way of doing this is as a reverse ray-tracing problem. An algorithm iterates through the possible paths a photon can take through the detector, and evaluates a Gaussian likelihood under the hypothesis of each particle type⁵. This

likelihood is built from the spatial and temporal residuals between the detected photon hits and the expected photon hits from the ray-tracing. For example, a Gaussian spatial likelihood can be expressed as:

$$\mathcal{L}_h = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \langle x_{i,h} \rangle)^2}{2\sigma^2}\right) \quad (7)$$

where x_i is the detected position of the i -th photon, $\langle x_{i,h} \rangle$ is the expected position of the i -th photon under hypothesis h , and σ is the spatial resolution of the detector system. The hypothesis h can be for example an incident pion or a kaon. Note that many algorithms choose to use non-Gaussian likelihoods. By calculating the likelihood for each hypothesis, we can then form a log-likelihood ratio to determine which hypothesis is more probable:

$$\Delta \log \mathcal{L} = \log \mathcal{L}_{\text{kaon}} - \log \mathcal{L}_{\text{pion}} \quad (8)$$

This can be minimised across all detector hits to find the most probable particle type using various numerical methods. Variations on this spatial method include time-imaging⁶, where precise photon timing information is also used to build probability density functions (PDFs) as a supplement to spatial information for each particle hypothesis. The detected photon times are then compared to these PDFs to form likelihoods. Many modern algorithms use a combination of spatial, temporal, and even angular information from other sub-detectors to build more robust likelihood functions.

Likelihood-based methods are among the most powerful and widely used analysis techniques in countless experiments. Their practical application, however, often requires detailed modelling of detector responses and the evaluation of high-dimensional likelihoods, leading to significant computational overhead in complex detectors. Machine learning (ML) approaches offer a promising alternative by replacing explicit likelihood construction with data-driven inference. Neural networks can be trained to learn the detector response to different particle hypotheses directly, thereby implicitly capturing the information normally encoded in likelihood functions. This has already seen encouraging results on Geant4-simulated data for the Hyper-Kaminokande experiment⁷ and for the PANDA Barrel DIRC⁸.

III. PHOTON DETECTION TECHNOLOGIES

III.A. MCP-PMTs

Micro channel plate photomultipliers (MCP-PMTs) are devices that convert and amplify incoming photons into a measurable electrical signal. Photons incident on a photocathode emit electrons through the photoelectric effect. Single electrons are difficult to measure, so the number of electrons is amplified using a MCP. These are thin wafers containing thousands of μm capillaries. Via secondary emission, more electrons than are incident are produced⁹. This amplified electron signal is then recorded on an anode plate as an analogue voltage.

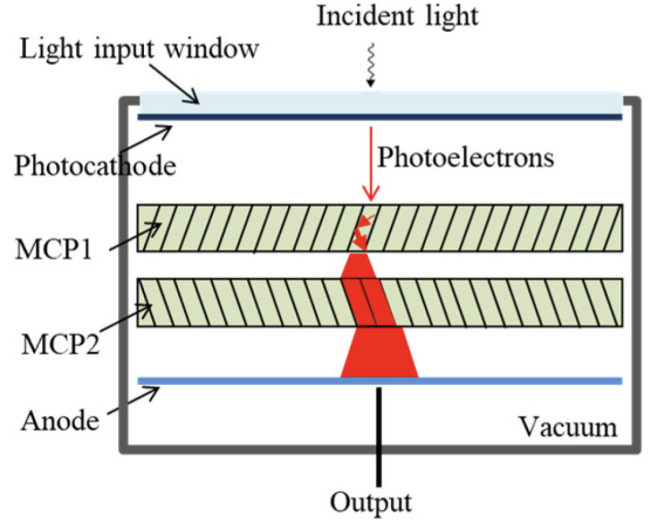


FIG. 3 **Cross section of a traditional MCP-PMT.** Due to the proximity of the anode and the photocathode, we are able to determine where the incident photon hit. MCP-PMTs are used in an incredible range of fields, from blood tests, to spectroscopy, to radar jamming¹⁰.

In detector physics we are interested in a few properties of MCP-PMTs. Firstly, the quantum efficiency (QE) of the photocathode is the probability of an incident photon producing a photoelectron; around 20% is typical.

The consistency of accurate hits is crucial to Cherenkov angle-imaging systems. Uniform gain across the entire MCP-PMT is ideal. Also, a higher gain means higher peaked counts, and therefore less uncertainty in the timing of the count. We must also minimise faulty signals from dark counts. Dark counts are spurious signals caused without any incident photon, often due to thermionic emission or field emission. Balancing the gain behaviour, determined mostly by the anode bias, with minimising dark counts is a key challenge. Afterpulsing is where positive ions are created during the electron avalanche. These positive ions accelerate backwards, towards the photocathode, causing potential damage and liberating secondary electrons. This creates a correlated, slightly delayed signal, appearing as a false photon hit.

The timing resolution of the MCP-PMT refers to the width of the response of the sensor from a single photon hit. Modern devices are well below 50ps single photon time resolution^{11–13}. A lower timing resolution allows us to be more precise with event reconstruction in detectors that rely on timing information for PID, such as DIRC systems.

Lastly, the lifetime of the MCP-PMT is of interest. The main factors affecting lifetime include overall exposure to photons, radiation damage and heat degradation. Minimising afterpulsing is also important to prevent damage to the photocathode. MCP-PMT development is an extremely broad field spanning multiple disciplines; this overview just scratches the surface. The topic is central to PID and detector physics as a whole.

III.B. SiPMs

Silicon photomultipliers (SiPMs) have been gaining traction recently due to developments in materials, leading to a great ease of manufacturing. They are tiny compared to an MCP-PMT, due to their use of single photon avalanche diodes (SPADs), each containing an avalanche photodiode and a quenching resistor. The simulations by F. Acerbi and S. Gundacker¹⁴ provide an excellent in depth description of SiPMs.

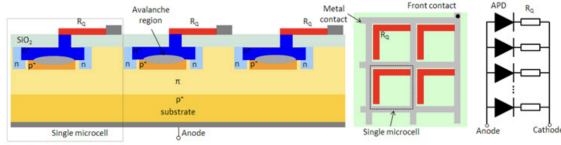


FIG. 4 **Cross section of SiPM.** SiPMs are externally biased such that each avalanche photodiode is at breakdown voltage, allowing for electron avalanches. The gain from a SiPM roughly matches that of a MCP-PMT at 10^5 – 10^6 times amplification¹⁵

SiPMs are much cheaper than MCP-PMTs and have a higher QE, however suffer from a high dark count rate and do not necessarily have similar hit rate capabilities, as the hit rate is mainly dependent on the density of SPADs. This has led to the explosion of SiPMs in fields like positron emission photography (PET) where precise photon timings are not necessary, and where ruggedness and sturdiness is required. SiPMs also operate at a far lower voltage than MCP-PMTs and are not affected by magnetic fields, adding to their dependability in medical settings.

The high dark count rate degrades PID in extremely precise systems, as noise causes statistical event reconstruction methods to struggle. Also, the dead time of SiPMs poses thought-provoking problems systems which rely on photon arrival time for strong PID performance. In some detector systems, a few thousand photons are guided to a small detector area. These all arrive typically within around 20-50ns, depending on the length of the detector. The multiplicity of a small area (pixel) on the photosensor array can often be ≥ 10 photons. The typical dead time of one SPAD is between 10–50ns¹⁶, so even though the microcells are much smaller than an MCP-PMT pixel, in the particularly photon-dense areas of the detector, it is possible to entirely lose some photons due to the dead time of the SiPM.

SiPMs provide an interesting alternative to MCP-PMTs, where their low cost and robust characteristics make them an extremely lucrative option for many applications in photon imaging. Their performance in high energy physics applications is certainly going to improve in the future, and they are already a serious consideration for future LHCb RICH upgrades¹⁷ despite being a relatively new technology (circa 2000) compared to PMTs (1930)¹⁸.

IV. CHERENKOV IMAGING DETECTORS

IV.A. RICH

Ring Imaging Cherenkov Detectors (RICH) are a class of Cherenkov detectors that use a radiator at a known distance from a photodetector array to measure the Cherenkov angle of incident particles. Shown in Fig 5 is a simplified Geant4 simulation of a RICH detector.

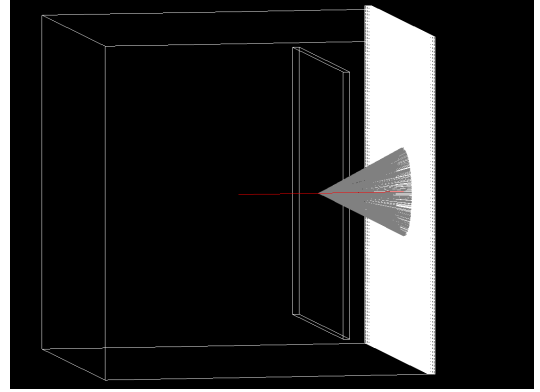


FIG. 5 **Geant4 simulation of the simplified RICH detector.** The particle track (red) is incident on some material with a refractive index n , called the radiator. In this case, an aerogel radiator is simulated. The resulting Cherenkov photons (gray) emitted in a cone are incident on a square array of sensitive photodetectors. In practice, the detector array is often a ring, and there are optics surrounding the device to guide photons towards the detector.

The Cherenkov angles are emitted in a cone around the particle track, necessitating the ring shape; having near light speed particles directly incident on photodetectors is likely to be expensive. The outer radius of the ring is often encased in mirrors or lenses to trap Cherenkov photons and guide them towards the photodetector array. These optical elements are carefully manufactured to minimise chromatic dispersion and to maintain angular resolution.

IV.A.1. Radiators

Different RICH experiments use different radiator materials. Common choices include aerogel such as in the Belle II ARICH¹⁹, gaseous radiators such as in the old LHCb RICH²⁰ and (although in mostly retired detectors) liquids such as in the barrel CRID at SLAC²¹.

Consider the Cherenkov angle condition from Eq 1. The radiator refractive index n is the key parameter that determines the choice of radiator, depending on expected beam momenta. Higher refractive indices are necessary for lower momentum particles, which is why we see older accelerator systems using liquid radiators with inherently higher refractive indices.

The refractive index is also wavelength dependent, which is a source of chromatic dispersion. This is an important consideration for radiator materials, as chromatic dispersion can degrade angular resolution. The

refractive index also affects the number of Cherenkov photons produced, as per the Frank-Tamm formula in Eq 3. There are also tricks to play with the radiator design to improve photon yield, such as layering aerogel with different refractive indices without performance cost²². Naturally though complexity comes at a cost, especially in the case of finely engineered components such as Cherenkov radiators.

IV.A.2. Optics

Optical elements are often used in RICH detectors to increase photon yield. Lenses can be used to focus Cherenkov photons onto smaller photodetector areas, reducing cost and size. Mirrors can be used to reflect photons that would otherwise escape the detector. Other optical tricks such as the aforementioned layered radiator (also called proxy focusing) can be used to tighten the spread of the photon cone, allowing lower momentum charged particles to be measured.

As an applied example, the pFRICH at the forward section of the ePIC detector makes use of conical mirrors to increase the acceptance angle of particle tracks which can be seen in Fig 6. More photons means better angular resolution and therefore improved event reconstruction.

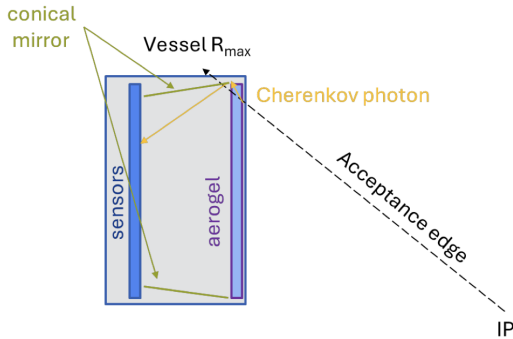


FIG. 6 **Schematic cross section of the pFRICH.** The conical mirrors (green) near the edge of the vessel mean that Cherenkov photons can be detected from steeper angles than would be traditionally possible²³

IV.A.3. Mechanics & Integration

Modern detectors are large, complicated systems that enjoy premium real estate. Integrating a Cherenkov detector into such an array presents a unique set of mechanical and design challenges. Mechanical tolerances and alignment are crucial to maintaining accuracy. Optical elements must be precisely aligned and consistently calibrated throughout the detector lifetime, requiring robust mechanical support and space-hungry laser calibration systems. Careful thermal and radiation management are also necessary to ensure the longevity of photosensitive components. Consideration of the surrounding magnetic field from the solenoid used by the TOF detectors is

also necessary, especially for PMTs which are magneto-sensitive.

Ensuring that the detector can be serviced and maintained is also a key consideration, especially for large scale experiments that may run for decades. Some experiments are built on rails to allow the inner detectors (RICH, TOF, trackers etc.) to be slid out for maintenance. These mechanical and integration challenges are often underappreciated, but are imperative to the success of a detector.

IV.B. DIRC

Devices that detect internally reflected Cherenkov light are a type of Cherenkov imaging detector are called DIRCs for short. They circumvent the need for a separate radiator by using solid transparent bars, typically quartz, to both generate and guide Cherenkov photons to the photodetectors. Fig 7 shows a detailed rendering of the PANDA barrel detector²⁴.

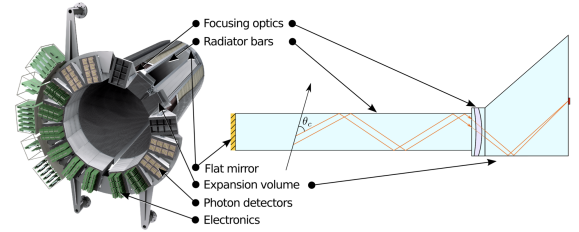


FIG. 7 **The PANDA Barrel DIRC along with an individual bar cross section.** 16 individual DIRC detectors are laid in a barrel-like arrangement, forming the very core of the detector array. Photons are internally reflected through the quartz bars.

The basic principle of DIRC operation is similar to that of other Cherenkov detectors; charged particles incident on a medium with refractive index $n > 1$ emit Cherenkov photons in a cone, and we try to measure the principle angle of this cone. However, in a DIRC detector the complexity arises from the optics of trapping and guiding the Cherenkov photons to a small array of photodetectors.

IV.B.1. Photon Transport via Internal Reflection

Similarly to photon transport in optical fibres, the Cherenkov photons produced in the quartz bars of a DIRC are trapped via total internal reflection as shown in Fig 7. The photons propagate to the end of the bar, where they are focused into an expansion volume, and then onto the photodetector array. The DIRC relies on photons internally reflecting hundreds of times with minimal loss, all while preserving the Cherenkov angle precision. The fused silica bars provide extremely smooth surfaces to facilitate this, with surface roughness possible on the order of angstroms²⁵.

The quartz bars are also UV transparent, allowing Cherenkov photons in the near-UV range to be detected,

increasing the photon yield of the detector. Quartz is also known to be radiation hard, which will allow the detector to operate in the high radiation environments of modern accelerators without significant degradation over time²⁶. The fused silica bars also have a near optimal refractive index of around 1.47 in the visible spectrum, giving large Cherenkov angles. Lower refractive indices would reduce the angle separation, harming PID performance while higher refractive indices would increase chromatic dispersion, harming angular resolution.

IV.B.2. Chromatic Effects & Timing Resolution

Since Cherenkov photons are produced across a range of wavelengths, chromatic dispersion is unavoidable. Since the quartz bars are often meters long, blue photons often arrive 100s of picoseconds later than red photons. This is due to the wavelength dependence of the refractive index, which is around 1.50 for blue light and 1.47 for red light in quartz. However, dispersion can be corrected for if the photon arrival time is measured with sufficient precision, making it a crucial aspect of DIRC design.

Modern MCP-PMTs are able to achieve single photon timing resolutions on the order of 15–20ps^{11–13}. Improvements in readout electronics and signal processing are also pushing timing resolutions lower²⁷. These advancements allow photon timing to become a measurable observable, rather than just noise; detected photons provide x, y, t rather than just x, y . We can bake this information into our event reconstruction algorithms, simulating expected photon arrival times for bluer and redder photons separately, massively improving angular resolution.

These sorts of chromatic corrections via timing information lead to far more sophisticated event reconstruction than RICH systems. As discussed in Section II.C, statistical methods such as maximum likelihood estimation are extremely powerful. DIRC systems offer a unique double-edged sword; complex photon paths, internal reflections, and chromatic dispersion make analytical solutions near impossible, but the extremely detailed information available from precise timing measurements allow for robust statistical models such as time-imaging⁶ and ML to shine.

IV.B.3. Reconstruction Methods & PID

While DIRC systems provide excellent PID performance in compact geometries, their reliance on time-based global likelihood reconstruction from chromatic effects can complicate hit-track association in high-multiplicity collider environments compared to spatially imaging RICH detectors.

The aforementioned time-imaging reconstruction method builds up a PDF from expected photon arrival times for all detector array channels and particle types. Then, events to be classified can be compared to these PDFs on a maximum likelihood basis. The algorithm can be optimised using a lookup table of possible paths

a photon could take to a specific channel, eliminating a few unnecessary computations. This allows more flexibility in focusing optics within the detector geometry, as possible paths can be analytically determined and placed in the lookup table. This improves photon yield and therefore PID performance, up to $4.8 \pm 0.1\sigma^6$. However, from a software standpoint, the algorithm is unfriendly to parallelise due to heavy use of conditional logic. Parallel hardware prefers to perform the same operation on many different data points. This algorithm demands different operations depending on detector channel, timing, and hypothesis. As such, computation time becomes a problem for large datasets.

ML on the other hand is significantly easier to parallelise. Raw detector data is fed to the network on a per-event level and a sequence of fixed operations (mostly matrix multiplications) are performed. This inherently uses fewer CPU cycles per event when compared to time-imaging. The ML models are also easy to distribute onto modern hardware. Also since latency is predictable per event, as it is correlated to model size, data processing is unlikely to ‘choke’; we are able to process batches of events at a time and can reliably expect all events to finish processing at around the same time. This is in contrast to variable latency such as in time-imaging reconstruction, where often batches are waiting on one particularly difficult event to finish processing before starting the next batch. These advantages make ML scaleable in a way that maximum likelihood algorithms are not.

Many different neural network architectures are being researched. This refers to the order and manner of the sequence of fixed operations that are done by the network. Dense networks consist of purely matrix multiplications, yet are seeing promising results⁸. More complicated networks such as normalising flows are closer to previous likelihood methods, but offer exact likelihood evaluation without iterative maximisation, combining the known strong performance of the time-imaging reconstruction method with the compute advantages of ML methods²⁸.

IV.C. Neutrino Identification with Event Topology

While angle-imaging detectors such as DIRC can not directly detect neutrally charged neutrinos, events involving charged particles can be detected and reconstructed in a system known as a topological-imaging detector. Since neutrino interactions are rare and large in scale, the detector and radiator material must span an enormous volume. The simple solution to this is to use easy or natural materials such as water or ice. This presents a unique set of engineering challenges, such as waterproofing the detectors and synchronising timing across large distances. It is also very difficult to separate the weakly interacting neutrino events from other background. Different collaborations have taken unique approaches to these problems depending on their physics programmes.

IV.C.1. Water Cherenkov Detectors (Super-K)

Water Cherenkov Detectors (WCDs) use large volumes of ultra-pure water as the radiator material. The Super-Kamiokande (Super-K) experiment in Japan is a famous example of a WCD, using 50,000 tons of water surrounded by over 11,000 PMTs, buried under a kilometre of mountain to shield from natural radiation. The detector is used to study neutrinos at MeV-GeV energies from various sources, including the sun, atmosphere, supernovae, allowing us to study neutrino-specific phenomena²⁹, which before Super-K were largely unexplored.

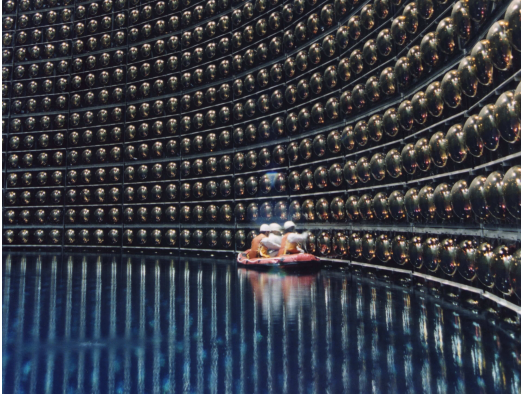


FIG. 8 Inside the cylindrical water tank of Super-K. Boats are used for the maintenance of water photodetectors. The detector is designed to identify neutrinos with MeV to GeV energies. (Photo: Kamioka Observatory, ICRR (Institute for Cosmic Ray Research), The University of Tokyo)

Super-K is squarely a precision detector for low energy neutrino physics. If we wish to study higher energy phenomena, such as cosmic neutrinos in the TeV-PeV range, we need to significantly increase the scale of the detector.

IV.C.2. Cosmogenic Neutrinos (IceCube)

The IceCube Neutrino Observatory is a cubic kilometre scale Cherenkov detector buried deep in the Antarctic ice sheet. It uses over 5,000 photodetector modules arranged on 86 strings, deployed between 1.5 and 2.5 km deep. The detector is designed to identify high energy neutrinos from astrophysical sources. These neutrinos are filtered from low energy neutrinos and other background by using the entire Earth as a shield; events that come from the core of the Earth are almost certainly high energy neutrinos, as other particles would be absorbed.

Fig 9 depicts high energy cosmic neutrinos being formed through the Greisen-Zatsepin-Kuzmin (GZK) process, where ultra high energy (≈ 8 joule) cosmic rays, accelerated by natural sources like black holes, interact with the cosmic microwave background radiation to produce pions, which then decay into neutrinos.

Fig 10 shows such a cosmic event from IceCube, where a high energy neutrino has interacted with the ice to produce muons and subsequent Cherenkov radiation. This figure is to give a sense of the scale of the detectors

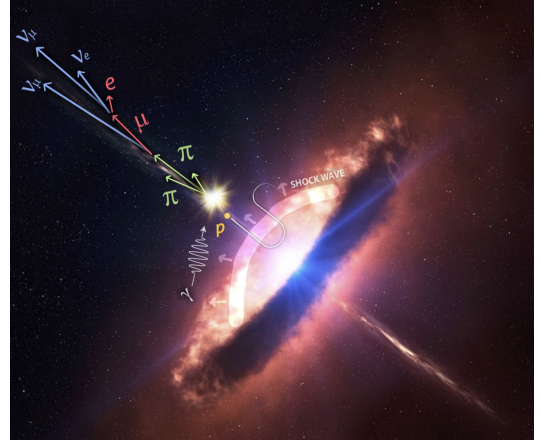


FIG. 9 Cosmic neutrinos formed through the GZK process. Protons accelerated by black holes are able to interact with the cosmic background, decaying through intermediary steps into various high energy neutrinos³⁰.

and the event topologies involved. The thickness of the strings represents photon intensity, while the colours represent timing information, red being at the beginning of an event and green/blue being later in time.

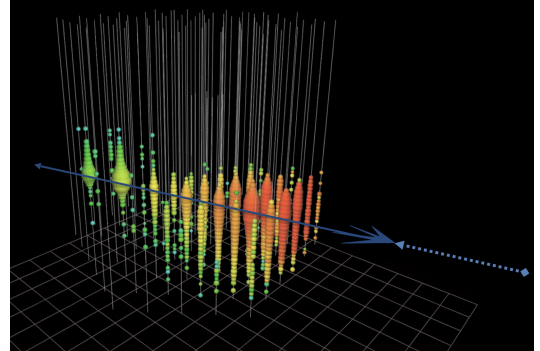


FIG. 10 IceCube event display. Multiple charged particle tracks can be distinguished. Maximum likelihood methods can be used to parse the muon tracks to identify neutrino interactions in the water³⁰.

Cherenkov detectors have proven to be invaluable tools in the current study of physics, but they also have a promising future. We will now look at some case studies of upcoming detectors.

V. CURRENT CHALLENGES & LIMITATIONS

Much of modern research is placed into improving photosensor performance in high-rate collider environments. Along with the new age SiPMs, improvements are also being made to MCPs, both towards better spatial and temporal resolution. A process known as atomic layer deposition (ALD) allows for extremely precise manufacturing of the capillaries in an MCP³¹, which have been shown to significantly improve sensor ageing³². Research into Large Area Picosecond Photodetectors (LAPPDs)

are attempting to reduce the cost per area of traditional MCP-PMTs while matching their performance characteristics. LAPPDs themselves have been shown to have good spatial and temporal resolution³³, and further characterisations of HRPPDs, a high-rate upgrade for LAPPDs, are ongoing^{34,35}.

On the software side developments being researched are as described in Sect IV.B.3 for DIRC systems. ML has the potential to offer fast particle classification in all Cherenkov imaging systems, either alone or in conjunction with other PID methods to decrease compute time. Effort is being focused into reducing model size, whilst retaining PID performance. This makes the models easier to deploy on the beamline and improves classification speed.

On the electronics side, developments are increasingly driven by the demand for precise timing and high-rate readout to fully exploit modern photosensors. Fast sampling ASICs³⁶ and high-bandwidth digitisation schemes³⁷ are required to preserve the picosecond scale timing resolution of MCP-based detectors. In high-rate environments, advances in front-end electronics and data acquisition systems aim to reduce dead time and manage large data volumes while maintaining low noise performance. Close integration of sensors with dedicated readout electronics is therefore an important area of research.

Finally, reducing the cost of detector systems will always be of great interest. The ease of manufacturing for SiPMs offer the potential to significantly reduce the price tag on detector arrays. Reuse of retired detector parts is also a potential cost saving measure; the recent transport of old BaBar boxes for use in the hpDIRC at Brookhaven is one example of successful reuse³⁸.

VI. CONCLUSION

This review introduced the field of Cherenkov imaging detectors, and covered modern developments towards precision PID in upcoming high-energy physics experiments. The detectors of today will be able to further our understanding of quantum chromodynamics, gluons in hadron events, and elusive high-energy neutrinos.

Highlights from the developments into the detectors include the push to improve spatial and timing resolutions of photosensors, with many difference devices reaching impressive single photon time resolutions. The lifetime and radiation resistance of the photosensors has also been developed, with novel machining techniques such as ALD. Furthermore, ML for particle classification is one of the fastest moving areas of the field. They have the potential to reduce computation time compared to other statistical PID methods depending on the model size, and have been shown to match the accuracy of the old methods with large detailed models.

There is no shortage of improvements to be made to these detectors. As physics programmes demand higher luminosities and higher energies, the technology must

adapt. There are clear limitations in the high-rate capability of photosensors and their DAQ systems. The size of data per second expected to be generated by near future detectors is immense. The cost of scientific programmes is ever more scrutinised, and so refinements towards the manufacturing of detector components are an area of particular focus.

Nevertheless, it is an exciting time for detector physicists; the FAIR international accelerator facility and the HL LHC upgrades are both due to complete construction before 2035. The Electron-Ion Collider at BNL is due to begin operations after 2035. Hyper-K, the successor to Super-K in Japan, is a neutrino detector that began civil engineering works and excavation 2 years ago.

Cherenkov imaging detectors have and will continue to be instrumental in high-energy physics research. They are a tried and true method of identifying charged particles. In the modern era, the detectors are beginning to move beyond auxiliary devices for separate mass hypotheses. Rather, they are playing vital roles in entire event reconstruction, allowing physicists to better study the interactions of our universe.

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